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Artificial intelligence in water quality monitoring: A review of water quality assessment applications

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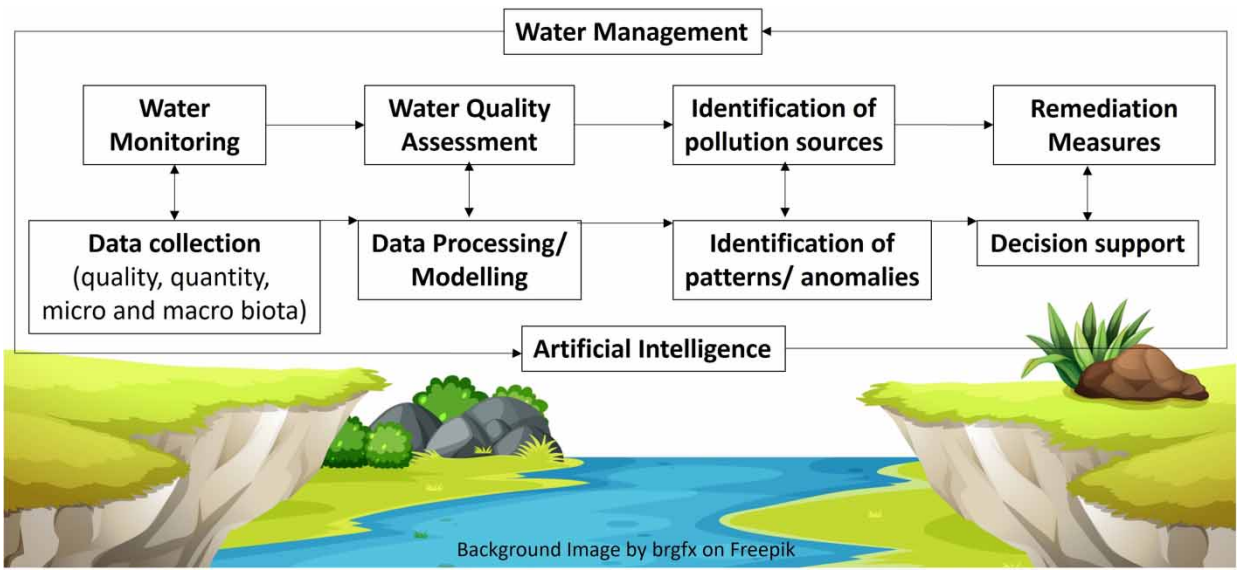
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ABSTRACT

Artificial intelligence (AI) has become a useful tool in numerous domains, including environmental science. This review explores the application of machine learning and deep learning, as AI technologies, applied in calculating and modelling water quality indexes (WQIs) and water quality classification. WQIs are used to assess the overall status of water bodies and compliance with environmental regulations. Given a large amount of monitoring data, traditional methods for calculating WQIs can be labour-intensive and subject to human error. AI offers a compelling alternative, with the potential to enhance accuracy, reduce time, and provide insights into complex environmental data. This paper examines recent progress in applying AI to water quality assessment through WQIs, including the creation of predictive models that incorporate diverse water quality parameters and the implementation of AI in real-time monitoring systems. The challenges of deploying AI, such as data availability, model transparency, and system integration, are also discussed. Through a detailed analysis of recent studies and practical implementations, this review analyses the potential of AI to contribute to water quality management and suggests directions for future research.

Key words: artificial intelligence, machine learning, water quality index

GRAPHICAL ABSTRACT



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INTRODUCTION

Water quality monitoring is an important responsibility of each state, to make sure that the population has access to safe water and the needs are met without creating pressures on the water resources. UN Sustainable Development Goal 6 aims to ensure the availability and sustainable management of water and sanitation for all. It focuses on providing safe and affordable drinking water, access to proper sanitation facilities, and promoting good hygiene practices. The goal also emphasizes the importance of protecting and restoring water ecosystems to maintain water quality and encourages efficient and sustainable management of water resources.

Water management is very complex and includes monitoring water quality and quantity parameters as well as biodiversity and aquatic life, identifying pollution sources and pollutants removal, sanitation, flood protection, resource allocation, etc.

The main sources of water pollution are discharges from urban agglomerations, leakage, and run-off from agriculture and industrial activities. Water monitoring includes a large number of hydrological, physical, chemical, and biological parameters, some of which are measured on site and others by sample analysis in the laboratory. Most countries have monitoring programs that specify sampling locations, parameters to be determined and frequency of sampling that are associated with efforts and costs for labour, reagents, equipment, etc.

Each country or region has its own quality standards that define limit values for parameters and classification systems to evaluate the state of a water body and its adequacy for different uses. For example, in the European Union, this field is regulated by the Water Framework Directive (WFD) and monitoring data are reported to the European Environment Agency by all Member States. The classification of the water bodies follows the One Out – All Out principle, which means that the status is given by the ranking of the worst parameter. For example, a water body cannot have 'good' status if one parameter ranks as 'poor'. Other countries have developed classification systems based on water quality indexes (WQIs), which are dimensionless numbers that aggregate the values of several selected indicators.

There are different WQIs, but the process of calculating the WQI usually includes the following steps:

- Selection of relevant water quality parameters;
- Assigning a weight to each parameter;
- Calculation of sub-indexes (e.g. based on limit values for a certain quality class);
- Aggregation of sub-indexes into the WQI.

According to the value of the WQI, the water is then categorized into quality classes, depending on the calculation method and water uses.

The WQI method is laborious and has several limitations that may lead to inaccurate results, but for many years it has been a good instrument to assess the overall water quality and its long-term trends.

Artificial intelligence (AI) refers to the simulation of human intelligence processes by machines, particularly computer systems, which include learning, reasoning, self-correction, perception, and interaction. AI can analyse in a short period of time huge amounts of data, identify patterns or anomalies, calculate indicators, provide visual representations of data, etc., which can support the assessment of water quality, as well as identification of pollution sources and remediation measures.

The recent development of AI tools for assessing water quality has the potential to bring significant improvements in this sector. It may reduce monitoring efforts and costs and increase the accuracy of WQI prediction and water quality classification (WQC) in several ways, some of which are mentioned below:

- Advanced analysis of monitoring data, followed by calculation of WQI may indicate which parameters have the strongest influence in determining the WQI value and allow reducing the numbers of monitored parameters or their frequency;
- Complex modelling of WQI values may allow the prediction of WQI;
- AI algorithms can assign datasets into quality classes based on raw monitoring data without the need of calculating WQIs.

In addition, the combination of AI and remote sensing may be able in the future to replace traditional monitoring methods with satellite data and real-time, on-site sensors data (IoT – Internet of Things), reducing the cost and efforts of water quality monitoring.

The objective of this paper is to conduct a comprehensive review of recently published research utilizing AI techniques in practical WQI and WQC applications. By examining results from diverse geographical locations and datasets, this study aims to provide a detailed analysis of available tools, their applicability, limitations, and areas for future research.

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METHODS

The databases such as scholar.google.com, scopus.com, and sciencedirect.com have been searched for publications on 'artificial intelligence water quality index' and 'machine learning water quality index' for the period 2015–2024 using terminologies as indicated in Table 1. Ninety publications were retrieved and were further analysed in terms of addressing

Table 1 | Terminology used in reviewed papers (abbreviations in brackets)

Water types	Coastal Water (CW), Drinking Water (DW), Groundwater (GW), Irrigation Water (IW), Surface Water (SW), Wastewater Treatment Plant (WWTP)
Water quality parameters	Dissolved Oxygen (DO), Hydrogen Potential (pH), Total Dissolved Solids (TDS), Electrical Conductivity (EC), Total Hardness (TH), Calcium (Ca) ions, Sodium (Na) ions, Potassium (K), Magnesium (Mg) ions, Total Alkalinity (TA), Chemical Oxygen Demand (COD), Suspended Solids (SS), Temperature (T), Biological Oxygen Demand (BOD/BOD_5), Nitrates (NO_3), Nitrites (NO_2), Ammonium (NH_4), Ammonia (NH_3), Dissolved Inorganic Nitrogen (DIN), Total Organic Nitrogen (TON), Phosphates (PO_4), Total Phosphorus (TP), Sulphates (SO_4), Chloride (Cl), Bicarbonate (HCO_3), Turbidity (TU), Transparency (TR), Faecal Coliform (FC), Total Coliform Bacteria (TC), Salinity (SAL), Molybdate Reactive Phosphorus (MRP), Chlorophyll a (CHL), Fats, Oils, and Grease (FOG), Manganese (Mn), Arsenic (As), Nickel (Ni), Boron (B), Lead (Pb), Zinc (Zn), Fluoride (F), Iron (Fe), Chromium (Cr), Cadmium (Cd), Copper (Cu), Selenium (Se), Mercury (Hg), Cobalt (Co), Potential Salinity (PS), Sodium Adsorption Ratio (SAR), Exchangeable Sodium Percentage (ESP), Magnesium Adsorption Ratio (MAR), Anionic Surfactant (LAS), Kelly Index (KI), Sodium Residual Carbonate (RSC), Magnesium Hazard (MH), Prussiate (CN), Sulphide (S)
WQI calculation method	Department of Environment Water Quality Index (DOE-WQI), Weighted Arithmetic Water Quality Index (WA-WQI), Canada Council of Ministries of the Environment Water Quality Index (CCME-WQI), National Sanitation Foundation Water Quality Index (NSF-WQI), Oregon Water Quality Index (OWQI), Minimum Operator Index (MOI), Irish Water Quality Index (IEWQI), Groundwater Quality Index (GQI/GWQI), Raw Water Quality Index (RWQI), Raw Water Quality Index Fuzzy (RWQIF), Irrigation Water Quality Indices (IWQIs), British Colombia WQI (BCWQI), Vietnam Water Quality Index (VN_WQI), Entropy Water Quality Index (EWQI), Weighted Quadratic Mean (WQM) WQI, Santiago-Guadalajara River (SGR-WQI), Nemerow Pollution Index (NPI), Water Quality Index-Department of Environment (Malaysia) (WQI-JAS), WQI + Fuzzy Hierarchical Analysis Process of the Water Quality Index (FAHP-WQI), Fuzzy-GIS-based Groundwater Quality Index (FGQI), Log-Weighted Quadratic Mean (LWQM), Sinusoidal Weighted Mean (SWM), Scottish Research Development Department (SRDD) Index, West Java (WJ) Index, Log-Weighted Quadratic Mean (LQM), Sinusoidal Weighted Mean (SWM), Heavy Metal Pollution Index (HPI), Normalized Difference Water Index (NDWI), Automated Water Extraction Index with no shadows (AWEI-nsh)
Classification	Water Quality Classification (WQC)
AI tools	Adaptive Boosting (AdaBoost), Adaptive Neuro-Fuzzy Inference System (ANFIS), Additive Regression (AR), Artificial Intelligence (AI), Artificial Neural Networks (ANN), Back Propagation Neural Networks (BPNN), Bagging Classifier (BC), Bagged Tree Model (BTM), Bootstrap, CATBoost, Convolutional Neural Network (CNN), Cubist Regression Trees (CB), Decision Tree Regressor (DT), Deep Feed-Forward Neural Network (DFFNN), Deep Neural Network (DNN), Discriminant Analysis (DA), Elastic Net Regression (ENR), Empirical Predictive Modeling (EPM), Ensemble Trees (ET), Extra Trees Regression (ETR), Extreme Learning Machine (ELM), Factor Analysis (FA), Feed-Forward Neural Network (FFNN), Gaussian Naïve Bayes (GNB), Gaussian Process Regression (GPR), Generalized Additive Models (GAM), Gradient Boosting Regressor (GB)/Gradient Boosting Machine (GBM), Gradient Boosted Trees (GBT), Histogram-based Gradient Boosting (HGBM), Isolation Forest (IF), K-Nearest Neighbors (KNN) Regressor, Kernel Approximation Regression (KAR), Kernel Density Estimation (KDE), Lasso Regression (LR), Levenberg–Marquardt three-layer back propagation algorithm (LMBP), Light Gradient Boosting (LightGBM/LGB), Linear Discriminant Analysis (LDA), Linear Regression (LR), Locally Weighted Linear Regression (LWLR), Logistic Regression (LR), Long Short-Term Memory (LSTM), M5P Tree (M5P), Machine Learning (ML), Mamdani Fuzzy Logic (MFL), Monte Carlo Simulation (MCS) (for model uncertainty), Multilayer Perceptron (MLP), Multilinear Regression (MLR), Multinomial Logistic Regression (MNLR), Multivariate Adaptive Regression Splines (MARS), Naïve Bayes (NB), Neural Net (NN), Neural Network Ensemble (NNE), Partial Least Squares Regression (PLSR), Particle Swarm Optimization (PSO), Polynomial Regression (PR), Principal Components Regression (PCR), Radial Basis Function Neural Network (RBFNN), Random Forest (RF), Random Subspace (RSS), Recurrent Neural Networks (RNN), Reduced Error Pruning Tree (REPT), Regression Trees (RT), Ridge Regression (RR), Stepwise Regression (SW), Stochastic Gradient Descent (SGD), Support Vector Machines (SVM), Support Vector Regressor (SVR), Takagi-Sugeno Fuzzy Neural Network, Extreme Gradient Boosting XGBoost Regressor (XGBR), Wavelet De-noising Technique-Based Augmented Neuro-Fuzzy Inference System (WDT-ANFIS)

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the topic of the review, and the information regarding the WQI methodology and AI tools that were used in the research. Fifty-six original research articles were included in the review, as they were found to include sufficient and relevant information about the water quality datasets and the methods used for processing them. The papers were reviewed for:

- Location and type of study site;
- Reference period of monitoring data;
- WQI methodology and included parameters;
- AI tools used in data processing and results regarding their performance.

The steps involved in the application of machine learning (ML) tools in water quality assessment are presented schematically in Figure 1.

WQI prediction tools are based on regression, which means that data from the past are analysed to identify patterns and relationships between independent variables (water quality parameters) and dependent variables (WQI value) and to predict how values will evolve. For a model to be reliable, it needs to be built on sufficient data that covers a relevant period of time. As an example, if water monitoring data are available for a period of 10 years, the first seven years can be used for model training, and the last three years to verify if the WQI values predicted by the model are in line with those calculated from actual data.

WQC classification tools are ML models that assign a label to a dataset, for instance 'good water quality'. These models also need to be trained on real data and checked for accuracy.

A selection of AI tools and their application is presented in Table 2.

RESULTS AND DISCUSSION

The research papers examined in this study encompassed a wide variety of geographical locations, time-spans of data collection, WQI calculation methods, and AI tools employed for data processing.

Most of the studies are on monitoring data from India (23.2%), followed by Pakistan (12.5%) and China, Ireland, and Malaysia (8.9% each) (Figure 2). A large number of studies are for surface water (SW) (51.8%), 33.9% for groundwater (GW), 8.9% for coastal water (CW), 3.6% for irrigation water (IW), and 1.8% for drinking water (DW) (Figure 2).

Three main categories can be distinguished based on the purpose and methodology of the studies:

- Complex datasets, including a large number of parameters that are used to calculate WQIs, followed by analysis of the influence of individual parameters on the final results and comparison between results obtained with all data vs. smaller datasets that still allow accurate prediction of water quality, in order to reduce the number of variables.
- Different methodologies for WQI calculation applied to the same dataset in order to assess their performance and improve the calculation method.
- Different ML methods applied to the same dataset and same WQI methodology, followed by assessment of performance by comparing the results of the AI models with those given by real monitoring data.

For instance, in a study carried out on monitoring data from Thailand for the period 2016–2021, it has been possible to reduce the number of parameters for WQI calculation from 13 to four, using ANN and the Bootstrap method (Chawishborwornworng *et al.* 2024). A long-term study on lake water in Finland (1980–2023) has shown that the long short-term memory model was the least sensitive model when COD and TP were removed from inputs, compared with SVR, RF, and ANN, using

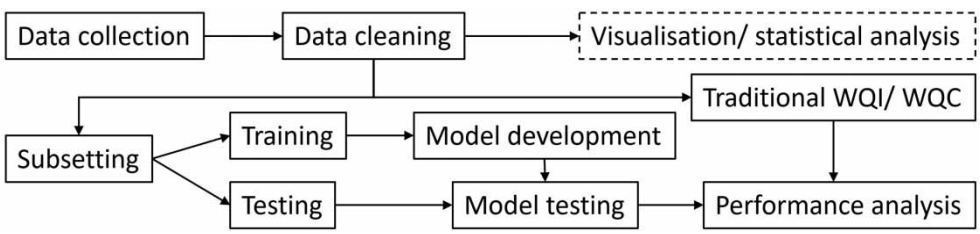


Figure 1 | General representation of ML application on water quality data (author's compilation).

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Table 2 | Selected AI tools and their possible application for WQI and/or WQC

AI tool	for	Features	Results
Random forest (RF)	WQI, WQC	Handles high-dimensional data well, robust against overfitting, and provides feature importance	Achieved high accuracy (Shams *et al.* 2023; Solangi *et al.* 2024)
Support vector machine (SVM)	WQI, WQC	Effective in high-dimensional spaces, robust against overfitting, especially with the right kernel	Demonstrated high accuracy in predicting WQI and classifying water quality (Haghiabi *et al.* 2018)
Artificial neural network (ANN)	WQI, individual parameters	Capable of capturing complex non-linear relationships, adaptable to various types of data	Effective in various studies with high accuracy in prediction tasks (Rana *et al.* 2023)
Long short-term memory (LSTM)	WQI	Excellent for time-series data, capable of learning long-term dependencies	Achieved high accuracy in studies often outperforming other models in time-series predictions (Nguyen *et al.* 2023)
Extreme gradient boosting (XGBoost)	WQI, WQC	High performance, efficient computation, and strong predictive power	Consistently high accuracy often outperforming other models (Solangi *et al.* 2024)
Decision tree classifier (DT)	WQC	Simple to interpret, handles both numerical and categorical data, useful for feature selection	High accuracy particularly effective in combination with ensemble methods (Solangi *et al.* 2024)
K-nearest neighbors (KNN)	WQC	Simple and intuitive, effective for small datasets	Good accuracy though performance can degrade with high-dimensional data (Zamri *et al.* 2022)
Adaptive neuro-fuzzy inference system (ANFIS)	WQI, WQC	Combines neural networks and fuzzy logic principles, effective for modeling complex relationships	Demonstrated good performance in various studies (Haghiabi *et al.* 2018)
CatBoost	WQC	Handles categorical data well, robust against overfitting, and efficient computation	Achieved high accuracy in studies often used in ensemble methods for improved performance (Nasir *et al.* 2022)
Multilayer perceptron (MLP)	WQI, WQC	Capable of learning complex patterns, adaptable to various types of data	Effective with high accuracy in prediction tasks (Palabıyık & Akkan 2024)
Logistic regression (LR)	WQC	Simple to implement, interpretable, and effective for binary classification problems	Good accuracy often used as a baseline model (Nallakaruppan *et al.* 2024)
Naive Bayes (NB)	WQC	Simple, fast, and effective for large datasets	Good accuracy, particularly effective for text classification and categorical data (Ilić *et al.* 2022)

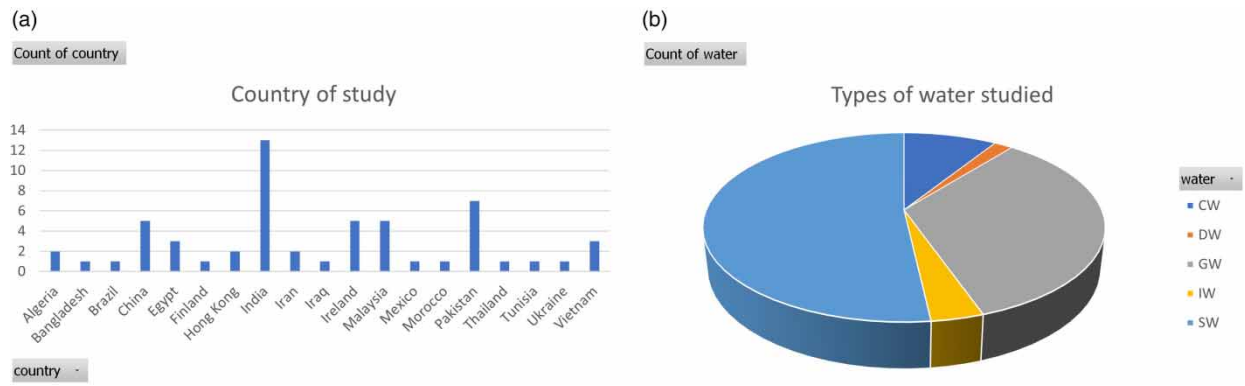


Figure 2 | Distribution of study areas (a) and types of water (b).

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the British Columbia WQI (Kim *et al.* 2024). Another study in China has allowed the reduction of the number of parameters from 22 to nine, using redundancy analysis (RDA), on a dataset from 2017 (Li *et al.* 2021). A study on irrigation water in Vietnam found that coliform, dissolved oxygen, turbidity, and total suspended solids are the most important parameters for water quality assessment (Lap *et al.* 2023). In another study, electrical conductivity had the highest influence on WQI for groundwater, while pH had the lowest influence (Raheja *et al.* 2022). However, there is also a report where 14 of 17 parameters included in WQI calculations were found to be significant. Moreover, the model was found to perform well also with only 12 parameters instead of 17 (Fernández Del Castillo *et al.* 2022). A noteworthy result is that, despite the superior performance of the 10-parameter model, a significantly simpler model incorporating only BOD, turbidity, and phosphate demonstrated a remarkably high level of accuracy in predicting river water quality (Asadollah *et al.* 2021).

Some of the reviewed papers aim to improve the water quality index methodology with the help of AI tools. The fuzzy-GIS-based groundwater quality index (FGQI), combining geographical information system (GIS) with the groundwater quality index, is proposed as a reliable tool for groundwater quality assessment (Jha *et al.* 2020). Another study made the first attempt to include heavy metal concentrations in groundwater assessment, demonstrating the robustness of the model (Sajib *et al.* 2023). The automated water extraction index with no shadows (AWEI-nsh) proposed by another group, using remote sensing data, may be applied when shadows are excluded from the image (Li *et al.* 2021). The WQI model could also be improved by adjusting parameter weight values and using new aggregation functions, namely sinusoidal weighted mean and log-weighted quadratic mean (Ding *et al.* 2023). In addition, using ML methods, satellite data can be used to improve water quality monitoring in coastal waters that are optically complex (Hafeez *et al.* 2019).

A comparison between models using a minimum number of parameters (coliform, pH, temperature, turbidity, and total dissolved solids) indicates that gradient boosting and polynomial regression performed better in predicting WQI, whereas MLP performed better in predicting WQC (Ahmed *et al.* 2019). However, it should be mentioned that the regression model only included a relatively short period of time (2009–2012). A study carried out by Abba *et al.* (2020) on a limited number of parameters indicated that the Adaptive Neuro-Fuzzy Inference System (ANFIS) model performed best at one location and BPNN at two other locations, which seems to indicate that the results may be specific to certain regions.

Two studies indicate that MLs are efficient tools to predict irrigation water quality and AdaBoost can predict all parameters (El Bilali *et al.* 2021; Trabelsi & Bel Hadj Ali 2022).

Several studies report successful application of regression models to predict WQI, but it is not clear how regression was possible for monitoring data collected during one year or even one set of samples collected from different locations (Kadam *et al.* 2019, 2022; Hussein *et al.* 2023; Ibrahim *et al.* 2023; Uddin *et al.* 2023b; Abbas *et al.* 2024). Aslam *et al.* (2022) used data for two years (2020–2021) and found that the hybrid RT-ANN algorithm can produce good results for short-term data, but the stability of the model would be improved using long-term datasets.

Both for regression models, as well as for classifiers, the research data show that there is no model that performs best for a majority of the datasets. In fact, almost each study reports a different method that gave the best results in their case, which seems to indicate that the outcomes depend on the AI tool, but also on the characteristics of the dataset and perhaps on the WQI or classification methodology. In each situation, there was either a different type of water or a different set of variables or WQI calculation method, so it is not possible to make comparisons between studies.

The AI tools reported by several studies to perform best in the case of WQI prediction are gradient boosting (Ahmed *et al.* 2019; Nguyen *et al.* 2023; Abbas *et al.* 2024), ANFIS (Tiwari *et al.* 2018; Hmoud Al-Adhaileh & Waselallah Alsaade 2021; Ibrahim *et al.* 2023), and SVM (Li *et al.* 2021; Shamsuddin *et al.* 2022; Ibrahim *et al.* 2023).

In the case of classifiers, several studies indicate good performance of XGBoost or other gradient boost method (Shams *et al.* 2023; Uddin *et al.* 2023b; Singh *et al.* 2024; Solangi *et al.* 2024), as well as SVM (Derdour *et al.* 2022; Hussein *et al.* 2023).

The review has revealed a diversity of datasets, water quality assessment methodologies and AI tools used for modelling, so this field appears to be very far from a world-wide standardization.

ML methods, including gradient boosting and polynomial regression, have shown effectiveness in predicting WQI, although results can vary based on the dataset and region. The research indicates that no single AI model consistently outperforms others across different datasets, suggesting that outcomes depend on the specific AI tool, dataset characteristics, and WQI methodology used. Overall, AI tools like gradient boosting, ANFIS, and SVM have been reported to perform well in WQI prediction, while classifiers like XGBoost and SVM have shown good performance in classification tasks.

The reviewed studies and the main results are presented in Table 3.

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Table 3 | Reviewed studies and their main results

Location/type of water[a]	Periods	WQI method/parameters	AI tools	Results	Observations/study cases	References
Ukraine/SW	2000–2021	WQI/BOD_5, SS, DO, NO_3, NO_2, SO_4, PO_4, Cl, NH_4	AAN, ELM, DTR, RF, BBA, GP, KNN, SVR, XGBR	GP, SVR, and XGBR 100% reliability ($R^2 = 1$)	Southern Bug River	Masood *et al.* (2023)
Ireland/CW	2019–2020	Eight WQI models/NH_3, TR, TON, T, DO, NH_4, pH, SAL, MRP, BOD, CHL	MCS, GPR	WQM and RMS models were found to exhibit a higher prediction accuracy	Cork Harbour coastal water	Uddin *et al.* (2023a)
China/GW	Not specified	WQI Uddin/pH, NH_3, Mn, Ni, B, Pb, Zn, F, COD, Fe	DT, RF, XGBR	XGBR model surpasses DT and RF models in water quality prediction	Groundwater Quality at the Yopurga Landfill	Zheng *et al.* (2024)
Malaysia/SW	2012–2018	DOE-WQI/DO, NH_3, BOD, COD, SS, pH	ANN, SVM, RF, NB	RF classifier outperformed NB, ANN, and SVM	Langat Basin in Selangor	Suwadi *et al.* (2022)
Pakistan/SW	2012–2019	WA-WQI, CCME-WQI, NSF-WQI, OWQI, MOI/pH, DO, EC, TU, FC, T	DT, KNN, LR, MLP, NB	DT algorithm had the highest classification accuracy of 99.6%	Rawal Dam, Islamabad	Ahmed *et al.* (2021)
Malaysia/SW	2001–2010	DOE-WQI custom/DO, BOD, COD, SS, NH_3, pH	RBFNN, BPNN	The BPNN model performed the best results of $R^2 = 0.7007$	Langat River and Klang River	Hameed *et al.* (2017)
Vietnam/IW	2005–2018	BOD_5, NH_4, PO_4, TU, TSS, TC, DO	XGBoost LSTM, RNN	Coefficients of determination ranging from 0.84 (RNN) to 0.96 (XGBoost)	Red River Delta irrigation water	Nguyen *et al.* (2023)
Malaysia/SW	2005–2014	DOE-WQI/DO, BOD, COD, NH_3, SS, pH	BPNN	CopulaGAN and TVAE outperformed other methods	Selangor River and Skudai River	Chia *et al.* (2023)
India/GW	2013–2014	GQI-10, GQI-7/TDS, NO_3, Ca, Mg, Na, Cl, K, F, SO_4, TH	Fuzzy-GIS-based Groundwater Quality Index (FGQI)	FGQI model can predict groundwater quality better than GQI-10 and GQI-7 models	Tiruchirappalli district, Tamil Nadu state in the southern part of India	Jha *et al.* (2020)
Ireland/CW	2017–2022	IEWQI/ pH, T, SAL, BOD_5, DO, TR, DIN, MRP, TON	LR, Regression Trees, SVM, GPR, KARs, ET, NN	IEWQI model is effective in evaluating the impact of various anthropogenic pressures	Cork Harbour	Uddin *et al.* (2023c)
Thailand/SW	2016–2021	pH, DO, T, BOD, FC, TSS, TDS, TC, TH, TU, TP, NO_3, NH_3	ANN Bootstrap	ANN had excellent performance compared with other models in terms of accuracy $R = 0.993$	Lower Mun River Basin	Chawishborwornworng *et al.* (2024)
Hong Kong/ SW	1999–2015	CHL, SS, TU/Remote sensing data	SVR, RF, ANN, CB, EPM	ANN exhibits the best performance $R = 0.9$, Machine learning methods outperformed the multivariate regression models	Pearl River Estuary	Hafeez *et al.* (2019)
Ireland/CW	2022	IEWQI/pH, DO, SAL, BOD_5, T, TR, TON, MRP, DIN	IF, KDE	R^2 increased from 0.92 to 0.95 when data outliers were removed	Cork Harbour	Uddin *et al.* (2024)
Algeria/GW	Not specified	EC, pH, Na, K, SO_4, NO_3, Ca, Mg, Cl, HCO_3	DT, KNN, DA, SVM, ET	SVM classifier obtained the highest forecast accuracy, with 95.4%	12 municipalities of the Wilaya of Naâma in Algeria	Derdour *et al.* (2022)
Iran/GW	2019	GQI, GWQI/a data-fusion index based on four pollutants: Mn, As, Pb, and Fe	Mamdani fuzzy logic (MFL), SVM, ANN, RF	RF ($R^2 = 0.995$) and MFL ($R^2 = 0.921$) had the best and worst performances, respectively	Gulfepe-Zarinabad sub-basin in northwest Iran, 28 groundwater samples	Nadiri *et al.* (2022)
Bangladesh/ GW	Not specified	GWQI/T, pH, EC, TDS, Zn, Fe, Mn, Cr, Cd, Cu	ET, GPR, LR, SVM, ANN, RT	The GWQI model had high sensitivity ($R^2 = 1.0$)	Savar sub-district of Bangladesh groundwater	Sajib *et al.* (2023)
India/SW	Not specified	DO, TC, BOD, NO_3, pH, EC	NN, RF, MNLR, SVM, BTM	MLR highest accuracy at 99.83%, SVM lowest accuracy at 96.98%	Datasets from the Kaggle website	Hassan *et al.* (2021)
Pakistan/SW	2009–2012	FC, pH, T, TU, TDS, NO_3	MLR, PR, RF, GB, SVM, RR, ENR, NN, MLP, GNB, LR, SGD, KNN, DT, BC	GB and PR performed better in predicting WQI, MLP performed better in predicting WQC	Rawal Water Lake	Ahmed *et al.* (2019)
India/GW	Not specified	TU, SO_4, TH, MGH, DO, BOD COD, NO_3, As	CNN, DNN, RNN	>95% accuracy based on R^2 values, RNN less precise	Gold mining sites of Kolar Gold Fields, Karnataka	Gupta *et al.* (2023)
Brazil/SW	2009–2014	RWQI, RWQIF CHL, FC, colour, *Cyanobacteria*, Fe, Mn, pH, TU	Fuzzy logic	The Spearman correlation coefficient between RWQI and RWQIF was 89%	24 water sources, associated with WWTP in the southeast of Brazil	Oliveira *et al.* (2019)
Egypt/GW	2020	IWQI/T, pH, EC, TDS, K, Na, Mg, Ca, Cl, SO_4, HCO_3, CO_3, NO_3	ANFIS, SVM	ANFIS and SVM achieved R^2 0.99 and 0.97 in training and 0.97 and 0.76 in testing	El Kharga Oasis, Western Desert of Egypt	Ibrahim *et al.* (2023)
China/SW	2016	pH, HCO_3, TP, TN, BOD, NH_3, Fe, Cu, Zn, volatile phenol, DO, TDS, Cl, SO_4, Na, Ca, Mg, COD, PO_4, Cr, remote sensing data	PSO + remote sensing spectral indices (difference index, DI; ratio index, RI; and normalized difference index, NDI)	The model based on RI, DI, and NDI values of the 1.6 order is much better than the others at predicting the water quality index of the study area ($R^2 = 0.92$)	In the Ebinur Lake Watershed, there are two prominent absorption features situated around 700 and 950 nm	Wang *et al.* (2017)

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China/SW	2012–2015	HPI/COD, BOD, NH_3, petroleum, TP, F, LAS, Pb, Cu, Zn, Se, As, Cd, Cr	Takagi-Sugeno fuzzy neural network	Ammonia nitrogen and total phosphorus were the main contaminants in the Huangshui River	The Huangshui River is a major tributary of the upper Yellow River	Zhao *et al.* (2022)
Tunisia/GW, IW	2019–2021	IWQ/TDS, PS, SAR, ESP, MAR, T, pH, EC	RF, SVR, ANN, AdaBoost	AdaBoost model is best for predicting all parameters (r 0.88–0.89)	Downstream Medjerda River Basin	Trabelsi & Bel Hadj Ali (2022)
Morocco/GW	2009–2019	IWQ/EC, pH, T, Cl, SO_4, CO_3, HCO_3, NO_3, NO_2, NH_4, Na, K, Ca, Mg	AdaBoost, SVR, RF, ANN	AdaBoost performed best, followed by RF	Berrechid Aquifer Groundwater	El Bilali *et al.* (2021)
India/SW	1999–2010	WQI/DO, pH, BOD, NH_3, T	BPNN, ANFIS, SVR, MLR, NNE	ANFIS was best for Nizamuddin station, while BPNN was best for Palla and Udi (Chambal) R (>0.9)	Nizamuddin, Palla and Udi (Chambal), across the Yamuna River, India	Abba *et al.* (2020)
Finland/SW	1980–2023	BCWQI/EC, DO, COD, pH, TP, TU, SD	SVR, RF, ANN, LSTM	LSTM is the least sensitive model to exclusion of COD and TP, $R = 0.91$	Lake Paijanne, Finland	Kim *et al.* (2024)
Egypt/GW	Not specified	GWQI/pH, EC, TDS, Na, SAR, PI, KI, RSC, MH	SVM	SVM WQI accuracy values ranging from 0.88 to 0.90	Abu-Sweir and Abu-Hammad, Ismalia, Egypt	Abu El-Magd *et al.* (2023)
Pakistan/GW	2022	WQI/EC, pH, TDS, HCO_3, Cl, SO_4, Ca, Mg, Na, K, NO_3, F, Fe, As	DT, SVM, KNN, ET, DA	SVM performed as best classifier, accuracy 90.8% for raw data and 89.2% for normalized data	Sakrand, province of Sindh	Hussein *et al.* (2023)
China/SW	2017	WQI/NH_3, COD, BOD_5, DO, TN, TP, TH, SS, Chroma, TU PO_4, Cr, SO_3, Fe, Cu, Zn, volatile phenol, Cl, Co, SAL, TDS pH	RF, SVM, PLSR, PLSR-SVM Sentinel-2MSI data at 10 m resolution NDWI	PLSR-SVM provided better WQI than the other models R^2v = 0.87 TDS, COD, and TN are the most influential in WQI. Proposed AWEI-nsh		Li *et al.* (2021)
Malaysia/SW	2009–2010	T, EC, SAL, NO_3, TU, PO_4, Cl, K, Na, Mg, Fe, FC	ANFIS, RBF-ANN, MLP-ANN, WDT-ANFIS	WDT-ANFIS model predicted well all the parameters ($R^2 \geq 0.9$)	Johor River Basin	Najah Ahmed *et al.* (2019)
Pakistan/GW	2022	WQI/TDS, Na, K, Ca, Mg, HCO_3, SO_4, Cl, pH, EC, NO_3, well depth	RF, GB, SVM, XGBoost, KNN, DT	RF and GB lead with 95 and 96% accuracy, SVM 92%. KNN 84%, DT 77%	422 data samples from Mirpurkash	Abbas *et al.* (2024)
Pakistan/GW	Not specified	EC, pH, TDS, Ca, Mg, TH, Cl, NO_3, NO_2, SO_4	LR, DT, XGBoost, RF, KNN	DT and XGBoost achieve accuracies of 100%. RF 88%, KNN 75%, LR 50%	Pano Aqil city, Pakistan, Indus River	Solangi *et al.* (2024)
India/SW	2005–2014	WQI/DO, pH, EC, BOD, NO_3, FC, T, TC	ANFIS, FFNN, KNN	ANFIS accuracy 96.17% for predicting WQI, FFNN 100% accuracy for WQC	Different locations in India (1,679 samples from 666 different sources)	Hmoud Al-Adhaileh & Waselallah Alsaade (2021)
Pakistan/SW	Not specified	COD, TOC, NH_3, As, Ni, Zn, oil and grease	AdaBoost, KNN, GB, RF, SVR, BR	GB performed best $R^2 = 0.88$ training, $R^2 = 0.85$ testing	Aik-Stream, industrially polluted, 150 sites	Ejaz *et al.* (2024)
China/SW	2021	WQI/DO, COD, BOD_5, NH_3, TP, TN, F, CN, S, Se, As, Cu, Zn, Hg, Cd, Cr, Pb, pH	LightGBM	Proposal on new aggregation functions and machine learning to improve water quality assessment	17 sampling sites in the Chaobai River Basin	Ding *et al.* (2023)
Ireland/CW	2019	Seven WQI models/T, pH, DO, TON, NH_4, MRP, BOD_5, TR, CHL, DIN	SVM, NB, RF, KNN, XGBoost	KNN (100% correct) and XGBoost (99.9% correct) were best for all seven WQI models	29 monitoring sites of the Cork Harbour	Uddin *et al.* (2023b)
India/SW	1996–2012	pH, EC, Cl, NO_3, NH_4, FC	Fuzzy C-means FCM-ANFIS and subtractive clustering SC-ANFIS	SC-ANFIS ($R^2 = 0.9919$) gave more accurate results than FCM-ANFIS	Eight different monitoring stations across River Satluj in northern India	Tiwari *et al.* (2018)
Algeria/GW	1999–2020	WQI/TDS, EC, T, pH, TH, Ca, Mg, Na, K, Cl, HCO_3, SO_4	MLR, RF, M5P, RSS, AR, ANN, SVR, LWLR	MLR model has higher accuracy in first scenario, RF better in second scenario $R = 0.9984$	Illizi region, southeast Algeria, 114 samples from 57 wells	Kouadri *et al.* (2021)
Vietnam/SW	2007–2020	VN_WQI/pH, TU, T, TSS, DO, BOD_5, NH_4, PO_4, COD, TC	MLR, SVM, DT, RF, MLP	The RF model delivers the highest accuracy in predicting the WQI, achieving a similarity score of 0.94	An Kim Hai irrigation system, north of Vietnam	Lap *et al.* (2023)
India/GW	2016	EWQI and WQI/pH, EC, TH, Ca, Mg, Na, K, HCO_3, Cl, SO_4, NO_3, F	DNN, GBM, XGBoost	DNN outperformed in predicting both indices	Haryana state (India) 392 datasets	Raheja *et al.* (2022)
Egipt/IW	2020	IWQI/pH, EC, Ca, Mg, Na, K, HCO_3, Cl, SO_4	SVM, XGBoost, RF, SW, PCR, PLS	SW emerged as the best followed by PCR and PLS	Bahr El-Baqr, Egypt, 105 water samples	Mokhtar *et al.* (2022)
India/GW	2015	WQI/pH, EC, TDS, TH, Ca, Mg, Na, K, Cl, HCO_3, SO_4, NO_3, PO_4	ANN, MLR, LMBP in ANN	The precision level is high in the ANN model	Shivganga River basin, Ghat region, 34 well samples	Kadam *et al.* (2019)
India/GW	Not specified	WA-WQI/pH, TU, EC, TDS, pH, TH, Cl, F	ANN, SVM, RF, XGBoost, MLR	XGBoost model gave best results: training $R^2 = 0.969$, testing $R^2 = 0.987$	Ujjain city of Madhya Pradesh in India, 54 samples from the urban area	Mohseni *et al.* (2024)

(*Continued.*)

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Table 3 | Continued

Location/type of water[a]	Periods	WQI method/parameters	AI tools	Results	Observations/study cases	References
India/SW	2004–2014	WQI/pH, EC, DO, BOD, NO_3, TC	DT, RF, GBT, ANN, SVM for WQC	Average accuracy >80 GBT performed better than other models	Major rivers and their tributaries of India ($n = 3{,}595$)	Singh *et al.* (2024)
Hong Kong/SW	1998–2017	BOD, COD, DO, EC, NO_3, NO_2, PO_4, pH, T, TU	ETR, SVR, DTR	ETR model generally yields more accurate WQI predictions ($R^2test = 0.98$)	Lam Tsuen River, Tai Po city	Asadollah *et al.* (2021)
India/SW	2016–2020	WQI/ DO, T, pH, TH, Cl, TC, EC, SO_4, Na, PO_4, K, BOD, F, NO_3	MLP, RF, SVM, NB, DT	MLP regressor and MLP classifier outperform other models	Bhavani River, Kerala and Tamil Nadu, 10,560 data samples, 31 attributes	Nair & Vijaya (2022)
Ireland/CW	2020	WQM-WQI/t, TON, NH_3, NH_4, DO, pH, SAL, MRP, BOD, TR, CHL	RF, DT, KNN, XGBoost, ExT, SVM, LR, GNB	DT, ExT, and GXB could provide accurate and robust results in predicting WQIs	Cork Harbour 29 monitoring locations	Uddin *et al.* (2022)
Mexico/SW	2009–2021	SGR-WQI/Cd, Cr, BOD_5, DO, FC, F, FOG, Hg, NH_3, NO_3, Pb, pH, TSS, S, TDS, T, Zn	SLR, MLR, LRR, GAM, LR, LDA	GAM was better for WQI, LR for WQC	Santiago-Guadalajara River	Fernández Del Castillo *et al.* (2022)
Iraq/SW WWTP	2015–2019	NPI/BOD, COD, TSS, Cl, pH, SO_4, NO_3, PO_4	ANN, FA	A successful ANN model was built based on NPI where the R^2 was 0.965	North Rustumiyia WWTP, Diyala River	Mohammed & Al-Obaidi (2021)
Vietnam/SW	2010–2017	T, pH, DO, BOD, COD, TU, TSS, TC, NH4, PO_4, turbidity (TUR)	AdaBoost, GB, HGBM, LGB, XGBoost, DT, ET, RF, MLP, RBF, DFFNN, CNN	All 12 models were good at WQI prediction, XGBoost was best ($R^2 = 0.989$)	Four monitoring stations on La Buong River	Khoi *et al.* (2022)
India/DW	2005–2014	DO, pH, EC, BOD, NO_3, FC, TC	SVM, RF, LR, DT, CATBoost, XGBoost, MLP	CATBoost model was the most accurate classifier with 94.51%	1,679 samples, various Indian states	Nasir *et al.* (2022)
Malaysia/SW	2012–2016	WQI-JAS/DO, BOD, COD, SS, AN, pH	ANN, DT, SVM	SVM was the best performing model for predicting WQI	Langat River Basin 560 records, 14 monitoring stations	Shamsuddin *et al.* (2022)
Iran/GW	Not specified	WQI + FAHP-WQI/EC, SAR, PI, MAR, KR, PS, SO_4, Cl, Na, Mg, Ca, HCO_3, TDS, pH, FC	GEP, M5P, MARS	MARS is slightly more accurate than the M5P model for estimating WQI ($R = 1$ training, $R =$ 0.999 test)	96 deep wells in the Yazd-Ardakan Plain	Goodarzi *et al.* (2023)
Pakistan/GW	2020–2021	PKWQI/ pH, DO, TDS, ES, SAL, Cl, TH, SO_4, NO_3	RT, RF, M5P, REPT, BA, CVPS, RFC	The RT-ANN algorithm outperformed all other algorithms in terms of accuracy, with the highest RSQ value of 0.951	39 locations COD and BOD were not considered	Aslam *et al.* (2022)
India/SW	2005–2014	DO, pH, EC, BOD, NO_3, FC, TC	RF, XGBoost, GB, AdaBoost for WQC. KNN, DT, SVR, MLP for WQI	GB gave best results, with a classification accuracy of 99.50%. MLP was best for WQI $R^2 = 99.8\%$	Lakes and rivers in India	Shams *et al.* (2023)

[a]Water types: surface water (SW), groundwater (GW), coastal water (CW), irrigation water (IW), and drinking water (DW).

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CONCLUSIONS

There is a wide variety of AI tools available to support WQI and WQC prediction, and some of them have been demonstrated to have high accuracy and reliability. Choosing the appropriate AI tool for WQI modelling depends on the specific requirements of the problem, the nature of the dataset, and the desired performance metrics. Ensemble methods like random forest and XGBoost, as well as neural network-based approaches like ANN and LSTM, are particularly effective for complex water quality assessment tasks. Combining these models or using hybrid approaches can further enhance prediction accuracy and robustness (Shams *et al.* 2023).

It should be mentioned though that the review has revealed a wide variety of datasets and locations, and that not all results may be easily replicated. Although some short-term studies may be justified by the lack of long-term monitoring data, the methods should be tested before application on different datasets.

AI and ML tools offer advantages such as the ability to process vast amounts of information and handle complex, non-linear relationships in water quality data, deal with partial or missing data, and provide real-time or near-real-time predictions, which can be crucial for water quality monitoring and management.

To make use of these recently developed tools, water managers may develop gradual approaches to reduce costs and improve the efficiency of monitoring, including:

- Analysis of historical datasets to identify critical parameters, pollution hotspots, and water quality trends;
- Integration of the spatial dimension (GIS) and real-time monitoring systems (sensors);
- Set alerts for key parameters that indicate when threshold values are exceeded for quick intervention;
- Develop new models for data integration and/or WQC based on individual parameters' inputs.

Some AI tools that can be applied include linear and logistic regression, decision trees, random forests, support vector machines, neural networks, etc.

Open access to water quality monitoring data has allowed data scientists to make fast and significant progress in developing new tools for extracting relevant information from the data, which can further support authorities to identify pollution sources, design remediation measures, and assess their efficiency in improving water quality.

In general, a model trained on data specific to a certain region is only able to produce accurate results for that region. If more monitoring data would become available, further research could explore how models developed in one part of the world perform in other regions and even contribute to a global model. Combinations of traditional monitoring and real-time data from *in-situ* probes and remote sensing could lead to quick identification of pollution sources and allow early interventions for remediation. In addition, climate change considerations could improve the accuracy of the models.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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